Type MS · Name thorax and airway structures and the route of air

Answer 1 of 8 · Question 4 part a.i · 3 marks

2024 JUN · Paper 02 · find part a.i in this mark scheme extract

Question number					
4	(a)	(i)	Correct labels to: trachea; bronchus/bronchi; alveoli;		(3)
		(ii)	• many alveoli/large surface area; • short diffusion distance/ alveoli 1 cell thick/capillary 1 cell thick/ thin wall; • moist; • (dense/extensive/large) network of capillaries;		(4)
	(b)	(i)	any four from • reference to use of a nose clip; • reference to soda lime (to absorb carbon dioxide) • reference to sterilising mouth piece; • breathe in and out(normally) through a mouthpiece(connected to spirometer); • pen goes up and down (on a rotating drum);\ • measure peak to trough/ highest to lowest/top to bottom/ from the trace;		(4)
		(ii)	4600 ÷ 30 (x 100) = 1380; 4600 – 1380 = 3220 (cm^3); or 4600×0.7=3220 (cm^3);;	ecf: allow correct calculation from values given for mp1. Max 2 marks	(2)

Total for Question 4 = 13 marks

Type MS · Name thorax and airway structures and the route of air

Answer 2 of 8 · Question 6 part a · 1 mark

2020 JAN · Paper 01 · find part a in this mark scheme extract

Question number				
6	(a)	bronchus/bronchi (1)		1
	(b) (i)	• inhale and exhale/breathe in an out (1) • through mouthpiece (1) • floating air chamber rises/falls (1) • pen records a trace on (the rotating) graph paper/drum (1)		4
	(ii)	Do not share the mouthpiece with another person/disinfect mouthpiece before use by another person/after use (1)		1
	(iii)	Any two from: • airways /bronchi/bronchus constricted/obstructed/reduced diameter (1) • less/reduced airflow (1) • per unit time (1)		2
	(c) (i)	cannot be transmitted from one person to another/not contagious/not caused by a pathogen/bacteria/virus/microorganism (1)		1
	(ii)	9.5 - 7 = 2.5(1) 2.5 ÷ 7 x 100(1) 36(%) (1)	Full marks for correct final answer Allow 35.7 for final answer	3
	(iii)	people taking less exercise/ consuming too much fat/sugar/ taking in more energy than is being used/ increase in the number of people smoking/drinking more alcohol/obesity (1)	Allow other valid lifestyle factor	1

Total for Question 6 = 13 marks

Type MS · Name thorax and airway structures and the route of air

Answer 3 of 8 · Question 2 part a.iii · 1 mark

2022 JAN · Paper 02 · extract 1 of 2 · find part a.iii in this mark scheme extract

Number				
2	(a) (i)	an answer that makes reference to three of the following: • warms air; • moistens air; • layer of (sticky) mucus; • removes/traps, pathogens/bacteria/microorganisms/dust / dirt (from entering lungs); • reference to 1st line of defence;	Mp3 must be linked to mp4 for mark	3
	(ii)	• prevents collapse; • to allow air to flow freely (to lungs/alveoli); • allows neck to bend;		2
	(iii)	B; (bronchi)		1
	(b)	D; (medulla oblongata)		1
	(c)	$17 \times 0.50 = 8.5\ dm^3$; 4×8.5; $= 0.34\ dm^3$;		3
	(d)	• reduced oxygen uptake by haemoglobin/forms carboxyhaemoglobin; • less oxygen to developing fetus; • lower rate of respiration/less energy released (by fetus); • less growth/lower birth mass;		4

Total 14 marks

Type MS · Name thorax and airway structures and the route of air

Answer 3 of 8 · Question 2 part a.iii · 1 mark

2022 JAN · Paper 02 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Name thorax and airway structures and the route of air

Answer 4 of 8 · Question 2 part a.i · 2 marks

2023 JUN · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

Number					
2	(a)	(i)	nose/nasal cavity/mouth, trachea/windpipe, bronchi, bronchioles;;	one mark for two correct two marks for four correct	2
		(ii)	• large surface area; • thin epithelium/walls;	R cell wall/cell membrane	2
	(b)	(i)	B; 0.04% A, C and D are all incorrect figures		1
		(ii)	• 1200×16; • $\frac{19200}{1000}$ = 19 dm^3;	ECF if any value used from table Accept 19.2	2
		(iii)	• $\frac{20 \times 10\,500;}{100}$ • $\frac{2100 \times 20;}{100}$ • = 420 cm^3;	2 mark ECF	3
		(iv)	• more carbon dioxide in blood/lungs/body; • detected by (chemo)receptors (in arteries); • rate of breathing/tidal volume increases; • increases to remove carbon dioxide;		4

Total 14 marks

Type MS · Name thorax and airway structures and the route of air

Answer 4 of 8 · Question 2 part a.i · 2 marks

2023 JUN · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Name thorax and airway structures and the route of air

Answer 5 of 8 · Question 5 part a.i · 3 marks

2025 JUN · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

5 (a) (i)		A = sternum; B = rib(cage); C = intercostal muscle;	ignore external/internal	3
	(ii)	• lungs; • diaphragm; • nose/trachea/bronchi/bronchus/bronchioles/alveoli;	in any order mark only first three	3
	(iii)	• diaphragm flattens; • ribs move up and out; • by contraction of intercostal muscles; • volume of thorax increases; • pressure in thorax/lungs decreases; • air forced/rushes/sucked/drawn, in (from outside);	reject moves down ignore external/internal ignore air flows in	6
(b)		• $C_6H_{12}O_6 + 6O_2$; • $6CO_2 + 6H_2O$; • energy/ATP;		3
				15 marks

Type MS · Name thorax and airway structures and the route of air

Answer 5 of 8 · Question 5 part a.i · 3 marks

2025 JUN · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Name thorax and airway structures and the route of air

Answer 6 of 8 · Question 5 part a.ii · 3 marks

2025 JUN · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

5 (a) (i)		A = sternum; B = rib(cage); C = intercostal muscle;	ignore external/internal	3
	(ii)	• lungs; • diaphragm; • nose/trachea/bronchi/bronchus/bronchioles/alveoli;	in any order mark only first three	3
	(iii)	• diaphragm flattens; • ribs move up and out; • by contraction of intercostal muscles; • volume of thorax increases; • pressure in thorax/lungs decreases; • air forced/rushes/sucked/drawn, in (from outside);	reject moves down ignore external/internal ignore air flows in	6
(b)		• $C_6H_{12}O_6 + 6O_2$; • $6CO_2 + 6H_2O$; • energy/ATP;		3
				15 marks

Type MS · Name thorax and airway structures and the route of air

Answer 6 of 8 · Question 5 part a.ii · 3 marks

2025 JUN · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Name thorax and airway structures and the route of air

Answer 7 of 8 · Question 4 part b.i · 1 mark

2025 JUN · Paper 02 (regional R) · find part b.i in this mark scheme extract

4	(a)	(i)	• decreases; • from 82% to 72%/by 10%		2
		(ii)	79:21; 3.76:1;	accept 3.7612:1 accept 3.762:1 accept 3.8/4:1	2
		(iii)	any two from • passive smoking/ex- smokers already have damaged lungs; • (air) pollution; • genetics/heredity;		2
		(iv)	tar;		1
	(b)	(i)	bronchiole;		1
		(ii)	alveolus;		1
		(iii)	any four from • damaged alveoli/alveoli walls broken; • less surface area (for gas exchange); • difficulty breathing/shortness of breath; • less oxygen passing into the blood/ less carbon dioxide removed from blood/less gaseous exchange; • reduced (aerobic) respiration; • fatigue/less energy/not very active;		4
				Total for Question 4 = 13 marks	

Type MS · Name thorax and airway structures and the route of air

Answer 8 of 8 · Question 4 part b.ii · 1 mark

2025 JUN · Paper 02 (regional R) · find part b.ii in this mark scheme extract

4	(a)	(i)	• decreases; • from 82% to 72%/by 10%		2
		(ii)	79:21; 3.76:1;	accept 3.7612:1 accept 3.762:1 accept 3.8/4:1	2
		(iii)	any two from • passive smoking/ex- smokers already have damaged lungs; • (air) pollution; • genetics/heredity;		2
		(iv)	tar;		1
	(b)	(i)	bronchiole;		1
		(ii)	alveolus;		1
		(iii)	any four from • damaged alveoli/alveoli walls broken; • less surface area (for gas exchange); • difficulty breathing/shortness of breath; • less oxygen passing into the blood/ less carbon dioxide removed from blood/less gaseous exchange; • reduced (aerobic) respiration; • fatigue/less energy/not very active;		4
				Total for Question 4 = 13 marks	